

FIITJEE –Phase Test (JEE-Advanced)

PHYSICS, CHEMISTRY & MATHEMATICS

Pattern - CPT-2

QP CODE:

PAPER - 2

Time Allotted: 3 Hours

Maximum Marks: 186

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Part.

- (i) **PART-A (01-08)** contains (8) Multiple Choice Questions which have **One or More Correct** answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: -1 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **-1 marks**, as a wrong option is also darkened.
- (ii) **Part-A (09-12)** – This section contains Two (02) List-Match Sets, each List-Match set has Two (02) Multiple Choice Questions. Each List-Match set has two lists: List-I and List-II. FOUR options are given in each Multiple Choice Question based On List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question. Each question carries **+3 Marks** for correct combination chosen and **-1 marks** for wrong options chosen.
- (iii) **Part-B (01-06)** contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

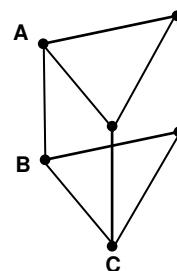
Enrolment Number : _____

BATCH – ABC LOT NAO PAPER-2123 (PHASE-5)

SECTION – I : PHYSICS**PART – A: (Multiple Correct Answers Type)**

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

- 10C of charge is given to a conducting spherical shell and another– 3C point charge is placed inside the shell. For this arrangement mark out the correct statement(s).
 (A) The charge on the inner surface of the shell will be +3C and it can be distributed uniformly or non uniformly
 (B) The charge on the inner surface of the shell will be +3C and its distribution would be uniform
 (C) The net charge on outer surface of the shell will be +7 C and its distribution can be uniform or non uniform
 (D) The net charge on outer surface of the shell will be +7C and its distribution would be uniform
- Nine similar resistors of resistance R are connected across edge of triangular pyramid as shown in the figure.
 (A) Equivalent resistance between points A and B is $\frac{3}{5} R$
 (B) Equivalent resistance between points A and B is $\frac{4}{3} R$
 (C) Equivalent resistance between points B and C is $\frac{9}{5} R$
 (D) Equivalent resistance between points B and C is $\frac{8}{15} R$
- Electric charge +q is uniformly distributed over the entire length of a ring of radius r rotating with constant angular velocity ω about its own axis. Assuming mass of ring to be m, its magnetic moment to be M and angular momentum to be L.
 (A) $M = \frac{q\omega r^2}{2}$
 (B) $M = \frac{2q\omega r^2}{5}$
 (C) $\frac{M}{L} = \frac{q}{2m}$
 (D) $\frac{M}{L} = \frac{2q}{m}$



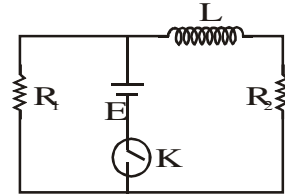
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4. In the L-R circuit shown in the figure, the key K is closed at time $t = 0$.

(A) the current through R_1 decreases with time, t
 (B) the drop across L increases with t

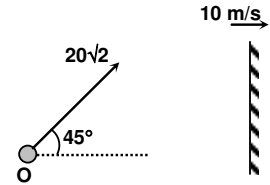
(C) the magnetic energy stored, in the steady state, equals $\frac{1}{2} L \frac{E^2}{R_2^2}$

(D) the total current through the battery is, in the steady state, $\frac{E}{R_1} + \frac{E}{R_2}$



5. A point object O is moving with velocity $20\sqrt{2}$ m/s and a plane mirror is also moving with a velocity 10 m/s as shown in the figure. Then

(A) the separation between the image and the mirror increases
 (B) the separation between the image and the mirror decreases
 (C) the velocity of image with respect to ground is 20 m/s and parallel to the mirror.
 (D) the velocity of image with respect to ground is 40 m/s and parallel to the mirror.

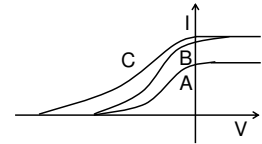


6. In the Young's double slit experiment, the interference pattern is found to have an intensity ratio between the bright and dark fringes as 9. This implies that

(A) the intensities at the screen due to the two slits are 5 units and 4 units respectively.
 (B) the intensities at the screen due to the two slits are 4 units and 1 unit respectively.
 (C) the amplitude ratio is 3.
 (D) the amplitude ratio is 2.

7. In a photoelectric experiment anode potential is plotted against plate current. Then which of the following is INCORRECT.

(A) A and B will have different intensities while B and C will have different frequencies.
 (B) B and C will have different intensities while A and C will have different frequencies.
 (C) A and B will have different intensities while A and C will have equal frequencies.
 (D) A and B will have equal intensities while B and C will have different frequencies



8. If the accelerating potential in an X-ray tube is increased, then the wavelengths of the characteristic X-rays may changes. Then which of the following is INCORRECT.

(A) first increases then decreases. (B) will increases
 (C) will decreases (D) remains constant

Space for rough work

PART – A: (Matrix-Match Type)

This section contains **TWO (02) List-Match Sets**. Each List-Match set has **TWO (02) Multiple Choice Questions**. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph

Four YDSE experiment 1, 2, 3 and 4 are performed using single monochromatic light of wavelength λ with separation between the slits d , $2d$, $3d$ and $4d$ respectively. The separation between the slits and screen is varying from D and $24D$. It is found that in experiment-I. Separation between slits is d , separation between slits and screen is D , fringe width is β and intensity at point P on the screen at distance of $x = \frac{D\lambda}{d}$ from central maxima is I_0 . Wavelength of light (λ) and position of point P(x) from central maxima in every experiment are same. List-I gives separation between slits while List-II lists the magnitude of some quantity.

List – I		List – II	
(I)	YDSE-1(d)	(P)	1
(II)	YDSE-2($2d$)	(Q)	$1/4$
(III)	YDSE-3($3d$)	(R)	$1/3$
(IV)	YDSE-4($4d$)	(S)	$1/2$
		(T)	$3/4$
		(U)	$2/3$

9. If the separation between slit and screen is D , then correct match for fringe width in terms of β is

Options

- (A) I $\rightarrow$ P, II $\rightarrow$ S, III $\rightarrow$ R, IV $\rightarrow$ Q
 (B) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ T
 (C) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ Q, IV $\rightarrow$ T
 (D) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ U

10. If the separation between slits and screen for experiment 1,2, 3 and 4 are kept fixed at D , $6D$, $12D$ and $24D$ respectively. Intensity at P on screen at a distance x from central maxima in terms of I_0 is

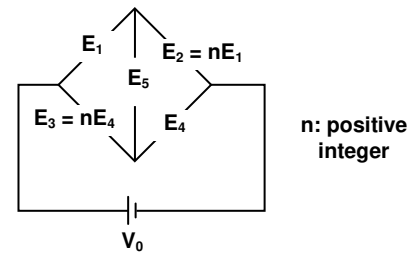
Options

- (A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ S, IV $\rightarrow$ U
 (B) I $\rightarrow$ P, II $\rightarrow$ U, III $\rightarrow$ S, IV $\rightarrow$ T
 (C) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ S
 (D) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ S, IV $\rightarrow$ T

Space for rough work

Answer the following by appropriately matching the list based on the information given in the paragraph

E_1, E_2, E_3, E_4, E_5 are five circuit elements such that E_1, E_2, E_3 and E_4 are of same nature (resistive or capacitive) and corresponding E_5 is capacitive or resistive respectively for a given experiment. Four value of $n(= 1, 2, 3, 4)$ give rise to four possible circuits: Circuit-1 ($n = 1$), circuit -2($n = 2$), circuit -3($n = 3$) and circuit -4 ($n = 4$). List I gives the possible four circuit and List II gives some numbers.



	List-I		List-II
(I)	Circuit-1	(P)	$\frac{4}{3}$
(II)	Circuit-2	(Q)	$\frac{3}{2}$
(III)	Circuit-3	(R)	1
(IV)	Circuit-4	(S)	$\frac{5}{2}$
		(T)	$\frac{8}{5}$
		(U)	2

11. If $E_1 = E_4 = R_0$ (resistive) and $E_5 = C_0$ (capacitive), then the correct match for the time constant of the voltage across the capacitor, as a factor of $R_0 C_0$, is:

Options

- (A) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ P
 (B) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ T
 (C) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S
 (D) I $\rightarrow$ R, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ T

12. If $E_1 = E_4 = C_0$ (capacitive) and $E_5 = R_0$ (resistive), then the correct match for the time constant of the voltage across the resistor, as a factor of $R_0 C_0$, is:

Options

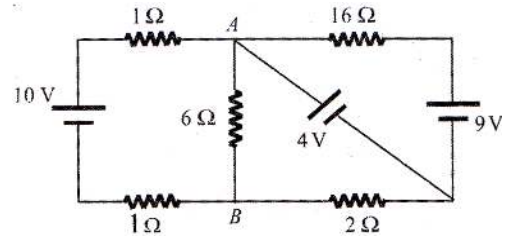
- (A) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ U, IV $\rightarrow$ S
 (B) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ U
 (C) I $\rightarrow$ R, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ T
 (D) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S

Space for rough work

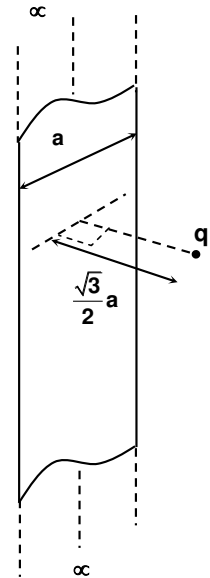
PART – B: (Integer Answer Type)

This section contains **06 questions**. Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking**.

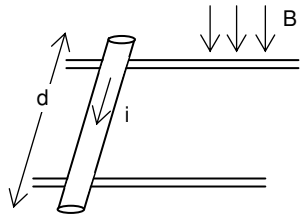
1. Find the potential difference (in Volts) between points A and B shown in the circuit.



2. There is an infinite long straight surface having width 'a'. A point charge q is placed at a perpendicular distance $\frac{\sqrt{3}}{2}a$ from the surface symmetrically as shown in the figure. If the flux linked with this infinitely long surface due to charge q is $\frac{q}{k\epsilon_0}$. Then find the value of k.



Space for rough work

3. A charged particle is projected in a magnetic field $\vec{B} = (x\hat{i} + 4\hat{j})10^{-2} \text{ T}$. The acceleration of the particle is found to be $\vec{a} = \left(\frac{8}{3}\hat{i} - 2\hat{j}\right) \text{ m/s}^2$. Find the value of x .
4. A cylindrical uniform rod of mass 0.72 kg and radius 6 cm rests on two parallel rails, that are $d = 50 \text{ cm}$ apart. The rod carries a current $I = 48 \text{ A}$ (In the direction shown) and rolls along the rails without slipping. If it starts from rest due to applications of uniform magnetic field of magnitudes 0.25 T (directed perpendicular to the rod and the rail), then the friction force (In N) between rod and rails is
- 
5. An electromagnetic wave with intensity 5 W/m^2 is absorbed by a surface of area A . If the force on the surface is 10^{-7} N , then A is (in m^2)
6. If the peak value of current through inductor (L), capacitor (C) and resistor (R) in parallel A.C. circuit are 2mA, 3 mA and 5 mA respectively. The peak value of current delivered by the source in the same circuit is $\sqrt{13k} \text{ mA}$. Find the value of K .

Space for rough work

SECTION – II : CHEMISTRY**PART – A: (Multiple Correct Answers Type)**

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

- Which of the following compounds have the oxidation number of the metals as +1?
 (A) CuSCN (B) $K_3[Cu(CN)_4]$
 (C) $[Fe(H_2O)_5NO]SO_4$ (D) Cu_2O
- Which of the following will undergo aldol condensation?
 (A) acetaldehyde (B) propanaldehyde
 (C) benzaldehyde (D) trideutero acetaldehyde
- For the reduction of NO_3^- ion in an aqueous solution, E^0 is +0.96 V. Values of E^0 for some metal ions are given below
 $V^{2+}(aq) + 2e^- \rightarrow V$ $E^0 = -1.19 V$; $Fe^{3+}(aq) + 3e^- \rightarrow Fe$ $E^0 = -0.04 V$
 $Au^{3+}(aq) + 3e^- \rightarrow Au$ $E^0 = +1.40 V$; $Hg^{2+}(aq) + 2e^- \rightarrow Hg$ $E^0 = +0.86 V$
 The pair(s) of metals that is(are) oxidized by NO_3^- in aqueous solution is(are)
 (A) V and Hg (B) Hg and Fe
 (C) Fe and Au (D) Fe and V
- The correct statement(s) regarding defects in solids is(are)
 (A) Frenkel defect is usually favoured by a very small difference in the sizes of cation and anion
 (B) Frenkel defect is a dislocation defect
 (C) Trapping of an electron in the lattice leads to the formation of F-center
 (D) Schottky defects have no effect on the physical properties of solids
- Extraction of copper from copper pyrite ($CuFeS_2$) involves
 (A) crushing followed by concentration of the ore by froth flotation
 (B) removal of iron as slag
 (C) self-reduction step to produce 'blister copper' following evolution of SO_2
 (D) refining of 'blister copper' by carbon reduction
- Mixture (s) showing positive deviation from Raoult's law at $35^\circ C$ is (are)
 (A) carbon tetrachloride + methanol (B) carbon disulphide + acetone
 (C) benzene + toluene (D) phenol + aniline
- The correct statement(s) pertaining to the adsorption of a gas on a solid surface is (are)
 (A) Adsorption is always exothermic
 (B) Physisorption may transform into chemisorption at high temperature
 (C) Physisorption increases with increasing temperature but chemisorption decreases with increasing temperature
 (D) Chemisorption is more exothermic than physisorption, however it is very slow due to higher energy of activation
- Which one of the following is NOT employed as a tranquilizer?
 (A) Naproxen (B) Tetracycline
 (C) Chlorpheniramine (D) Equanil

Space for rough work

PART – A: (Matrix-Match Type)

This section contains **TWO (02) List-Match Sets**. Each List-Match set has **TWO (02) Multiple Choice Questions**. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the lists based on the information given in the paragraph

List-I includes starting materials and reagents of selected chemical reactions. List-II gives structures of compounds that may be formed as intermediate products and/or final products from the reactions of List-I.

LIST-I		LIST-II	
(I)		(P)	
(II)		(Q)	
(III)		(R)	
(IV)		(S)	
		(T)	
		(U)	

9. Which of the following options has the correct combination considering List-I and List-II?

Options

(A) (II), (P), (S), (T)

(B) (II), (P), (S), (U)

(C) (I), (S), (Q), (R)

(D) (I), (Q), (T), (U)

10. Which of the following options has the correct combination considering List-I and List-II?

Options

(A) (IV), (Q), (U)

(B) (IV), (Q), (R)

(C) (III), (S), (R)

(D) (III), (T), (U)

Space for rough work

Answer the following by appropriately matching the lists based on the information given in the paragraph

Match the correct oxidation state(s) with the underlined element in the given compounds.

List – I (Compound)		List – II (Oxidation State)	
(I)	<u>Cr</u> O ₅	(P)	-2
(II)	H ₂ <u>S</u> O ₅	(Q)	+6
(III)	Na ₂ <u>S</u> ₂ O ₃	(R)	0
(IV)	CaO <u>Cl</u> ₂	(S)	+1
		(T)	+10
		(U)	-1

11. Which of the following options has the correct combination considering List-I and List-II?

Options

(A) (IV), (S), (U)

(B) (II), (P), (S)

(C) (I), (S), (Q)

(D) (I), (Q), (T)

12. Which of the following options has the correct combination considering List-I and List-II?

Options

(A) (IV), (Q), (U)

(B) (IV), (Q), (R)

(C) (III), (P), (Q)

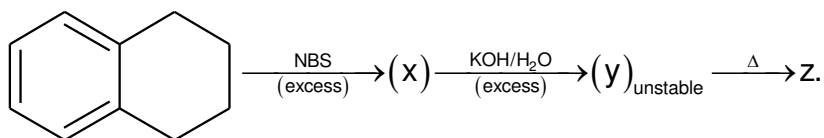
(D) (III), (T), (U)

PART – B: (Integer Answer Type)

This section contains **06 questions**. Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking**.

1. What will be the oxidation state of Cr in complex $[\text{Cr}(\text{C}_6\text{H}_6)_2]$?

2.

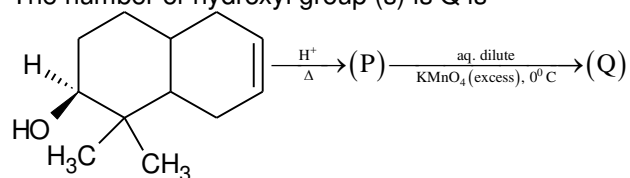


The sp^2 -hybridised carbon atoms present in one molecule of the end product z is:

3. How many equivalents of HCHO are formed by oxidative cleavage of one equivalent of D-glucose by excess of HIO_4 in water?

Space for rough work

4. The number of hydroxyl group (s) is Q is



5. A compound AB has a rock salt structure with $A : B = 1 : 1$. The formula weight of AB is $6.023 Y$ amu. And the closest distance between A and B is $Y^{1/3}$ nm. The observed density of the lattice in kg/m^3 is:
6. What is the molality of solution consisting of 33.0 mL of benzene (C_6H_6 ; $d = 0.877 \text{ g mL}^{-1}$) in 187 mL of hexane (C_6H_{14} ; $d = 0.660 \text{ g mL}^{-1}$)?

Space for rough work

SECTION – III : MATHEMATICS**PART – A: (Multi Correct Answer Type)**

This section contains **08 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

- If $AB = A$ and $BA = B$, then
 (A) $A^2 = A$ (B) $B^2 = B$
 (C) $A = I$ (D) $B = I$
- Let $R_n = |b_{ij}|_{n \times n}$ be a determinant such that $b_{ij} = \begin{cases} 1 & , |i - j| = 1 \\ 3 & , |i - j| = 0 \\ 0 & , \text{otherwise} \end{cases}$, then $n \in \mathbb{N}$
 (A) $R_{2019} + R_{2017} = 3R_{2018}$ (B) $R_{2019} + R_{2017} = 9R_{2018}$
 (C) The value of R_4 is 54 (D) R_5 is multiple of 9
- The area bounded between the curves $x^2 = y$ and $y^2 = x$ is equal to
 (A) area of the region $\{x, y\}: x^2 \leq y \leq |x|\}$ (B) $1/3$
 (C) $2 \int_0^1 (x - x^2) dx$ (D) none of these
- Given the differential equation $\frac{dy}{dx} = \frac{6x^2}{2y + \cos y}$; $y(1) = \pi$ and the following statements
 (A) solution is $y^2 - \sin y = -2x^3 + c$ (B) solution is $y^2 + \sin y = 2x^3 + c$
 (C) $c = \pi^2 - 2$ (D) $c = \pi^2 + 2$

Space for rough work

5. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are non-coplanar unit vectors also $\vec{b}$, $\vec{c}$ are non-collinear and $2\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \vec{c}$, then
 (A) angle between $\vec{a}$ and $\vec{c}$ is 60° (B) angle between $\vec{b}$ and $\vec{c}$ is 30°
 (C) angle between $\vec{a}$ and $\vec{b}$ is 120° (D) $\vec{b}$ is perpendicular to $\vec{c}$
6. If the line $\frac{x-2}{-1} = \frac{y+2}{1} = \frac{z+k^2-1}{4}$ is one of the angle bisector of the lines $\frac{x}{1} = \frac{y}{-2} = \frac{z}{3}$ and $\frac{x}{-2} = \frac{y}{3} = \frac{z}{1}$, then the value of k is / are
 (A) -3 (B) 2
 (C) 3 (D) -2
7. Let X and Y be two events such that $P(X) = \frac{1}{3}$, $P(X|Y) = \frac{1}{2}$ and $P(Y|X) = \frac{2}{5}$. Then
 (A) $P(X'|Y) = \frac{1}{2}$ (B) $P(X \cap Y) = \frac{1}{5}$
 (C) $P(X \cup Y) = \frac{2}{5}$ (D) $P(Y) = \frac{4}{15}$
8. If $I = \sum_{k=1}^{98} \int_k^{k+1} \frac{k+1}{x(x+1)} dx$, then
 (A) $I > \log_e 99$ (B) $I < \log_e 99$
 (C) $I < \frac{49}{50}$ (D) $I > \frac{49}{50}$

Space for rough work

PART – A: (Matrix-Match Type)

This section contains **TWO (02) List-Match Sets**. Each List-Match set has **TWO (02) Multiple Choice Questions**. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the lists based on the information given in the paragraph

Consider lines $L_1 : y = ax, z = c$; $L_2 : y = -ax, z = -c$ and $L_3 : y = z, ax = -c$ (where $a \neq b \neq c$). Let L be a variable line which intersects the given lines.

List-I and List-II contain statements about L and some and certain ordered triplets of numbers.

List – I		List – II	
(I)	Locus of L passes through point P then P is	(P)	$(a^2, 1, -c^2)$
(II)	Locus of L is $k_1x^2 - y^2 + k_2z^2 + k_3 = 0$, then (k_1, k_2, k_3) is	(Q)	$(0, 0, c)$
(III)	Locus L intersects the xz-plane in locus L' which is an ellipse with major axis α , minor axis β , centre as (α_1, β_1) then $(\alpha, \beta, \alpha_1 + \beta_1)$ is $(a^2 > 1)$	(R)	$\left(\frac{2c}{a}, 1, 0\right)$
(IV)	Locus L intersects the xy-plane in hyperbola with transverse axis α' , conjugate is β' , centre as (α'_1, β'_1) where $(\alpha', \beta', \alpha'_1, \beta'_1)$ is $\left(\frac{c^2}{a^2} > 1\right)$	(S)	$\left(2c, \frac{2c}{a}, 0\right)$
		(T)	$\left(\frac{c}{a}, 0, 0\right)$
		(U)	The locus of L and the xy-plane intersect in a hyperbola

9. Which of the following is the only **CORRECT** combination?
- A. (I), (Q), (T) B. (II), (P), (T)
 C. (III), (S), (R) D. (IV), (Q), (U)
10. Which of the following is the only **CORRECT** combination?
- A. (I), (Q), (R) B. (II), (P), (U)
 C. (III), (S), (R) D. (IV), (Q), (R)

Space for rough work

Let $f(x) = \sin(\pi \cos x)$ and $g(x) = \cos(2\pi \sin x)$ be two functions defined for $x > 0$. Define the following sets whose elements are written in the increasing order:

$$Z = \{x : g(x) = 0\}, W = \{x : g'(x) = 0\}$$

List – I		List – II	
(I)	X	(P)	$\supseteq \left\{ \frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, 7\pi \right\}$
(II)	Y	(Q)	an arithmetic progression
(III)	Z	(R)	NOT an arithmetic progression
(IV)	W	(S)	$\supseteq \left\{ \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6} \right\}$
		(T)	$\supseteq \left\{ \frac{\pi}{3}, \frac{2\pi}{3}, \pi \right\}$
		(U)	$\supseteq \left\{ \frac{\pi}{6}, \frac{3\pi}{4} \right\}$

11. Which of the following is the only CORRECT combination?
 A. (II), (R), (S) B. (I), (Q), (U)
 C. (II), (Q), (T) D. (I), (P), (R)
12. Which of the following is the only CORRECT combination?
 A. (IV), (P), (R), (S) B. (III), (P), (Q), (U)
 C. (IV), (Q), (T) D. (III), (R), (U)

Space for rough work

PART – B: (Integer Answer Type)

This section contains **06 questions**. Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer and **there will be no negative marking**.

- If, $z = \begin{bmatrix} O & I \\ I & O \end{bmatrix}$, where O, I are 2×2 null and identity matrix respectively, where $p = \det(\det(z))$ then, $|p|$ is _____
- Let the unit vectors $\vec{a}, \vec{b}, \vec{c}$ be the position vectors of the vertices of a triangle ABC. If $\vec{F}$ is the position vector of the mid point of the line segment joining its orthocentre and centroid then $(\vec{a} - \vec{F})^2 + (\vec{b} - \vec{F})^2 + (\vec{c} - \vec{F})^2 =$ _____
- The lines $\frac{x-1}{2} = \frac{y}{-1} = \frac{z}{2}$ and $x - y + z - 2 = 0 = \lambda x + 3z + 5$ are coplanar for λ , then $\left| \frac{7\lambda}{31} \right| =$ _____
- A bag contains 4 red and 6 black balls. A ball is drawn at random from the bag, its colour is observed and this ball along with two additional balls of the same colour are returned to the bag. If now a ball is drawn at random from the bag, then the probability that this drawn ball is red, is _____
- Three numbers are chosen at random without replacement from $\{1, 2, 3, \dots, 8\}$. The probability that their minimum is 3, given that their maximum is 6, is _____
- Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 3$ and $f'(0) = 1$. If $g(x) = \int_x^{\pi/2} [f'(t)\operatorname{cosec} t - \cot t \operatorname{cosec} t f(t)] dt$ for $x \in \left(0, \frac{\pi}{2}\right]$, then $\lim_{x \rightarrow 0} g(x)$ is equal to _____

Space for rough work

FIITJEE INTERNAL TEST

PHYSICS, CHEMISTRY & MATHEMATICS

CPT-2

QP CODE:

PAPER-2

ANSWERS

PHYSICS (SECTION -I)

PART- A

- | | | | |
|---------|---------|------------|------------|
| 1. A, D | 2. A, D | 3. A, C | 4. C, D |
| 5. B, C | 6. B, D | 7. B, C, D | 8. A, B, C |
| 9. A | 10. D | 11. D | 12. A |

PART - C

- | | | | |
|------|------|------|------|
| 1. 6 | 2. 6 | 3. 3 | 4. 2 |
| 5. 6 | 6. 2 | | |

CHEMISTRY (SECTION -II)

PART - A

- | | | | |
|---------------|------------|------------|------------|
| 1. A, B, C, D | 2. A, B, D | 3. A, B, D | 4. B, C |
| 5. A, B, C | 6. A, B | 7. A, B, D | 8. A, B, C |
| 9. B | 10. B | 11. A | 12. C |

PART-B

- | | | | |
|------|------|------|------|
| 1. 0 | 2. 8 | 3. 1 | 4. 4 |
| 5. 5 | 6. 3 | | |

MATHEMATICS (SECTION -III)

PART - A

- | | | | |
|---------|---------|------------|---------|
| 1. A, B | 2. A, D | 3. A, B, C | 4. B, C |
| 5. A, C | 6. A, C | 7. A, D | 8. B, D |
| 9. A | 10. B | 11. C | 12. A |

PART - B

- | | | | |
|---------|---------|---------|---------|
| 1. 1.00 | 2. 3.00 | 3. 1.00 | 4. 0.40 |
| 5. 0.20 | 6. 2.00 | | |

SOLUTION

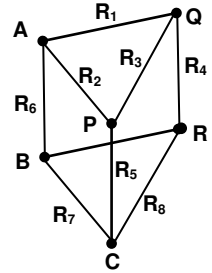
PHYSICS

PART – A

1. $-3C$ inside not at center (not fitted)
2. $V_P = V_Q$
Neglect R_3
 $V_R = V_C$
Neglect R_9
(R_2, R_5, R_7) and (R_1, R_4, R_8) sets are in series

$$\frac{1}{R_{eq}} = \frac{1}{3R} + \frac{1}{3R} + \frac{1}{R} = \frac{5}{3R}$$

$$R_{AB} = \frac{3R}{5} \Omega$$



3. $I = \frac{q}{T}$ $\therefore T = \frac{2\pi}{\omega}$

$$I = \frac{q\omega}{2\pi}$$

$$L = I\omega$$

$$L = mr^2\omega$$

$$\frac{M}{L} = \frac{q\omega r^2}{2mr^2\omega}$$

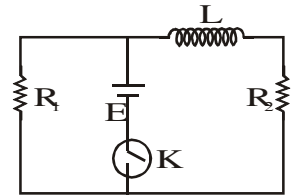
$$\frac{M}{L} = \frac{q}{2m}$$

$$M = IA = \frac{q\omega}{2\pi} \times \pi r^2 = \frac{q\omega r^2}{2}$$

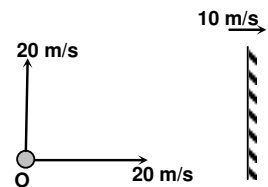
4. $i = i_0 e^{-t/\tau} = \frac{i_0}{e^{t/\tau}}$, $t \uparrow, i \downarrow$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$U = \frac{1}{2} L i^2 = \frac{1}{2} L \left(\frac{E}{R_2} \right)^2 = \frac{1}{2} L \frac{E^2}{R_2^2}$$



5. Image and mirror have relative velocity ($V_i = 0, V_m = 10 \text{ m/s}$)
 $V_i = 2V_m - V_0$ for perpendicular
 $V_i = 2 \times 10 - 20 = 0$
Image only have velocity parallel to mirror



6. $\frac{I_{max}}{I_{min}} = \frac{9}{1} = \frac{(x+1)^2}{(x-1)^2} \Rightarrow \frac{\left(\sqrt{\frac{I_1}{I_2}} + 1 \right)^2}{\left(\sqrt{\frac{I_1}{I_2}} - 1 \right)^2}$

$$\sqrt{\frac{I_1}{I_2}} = x, \frac{x+1}{x-1} = 3$$

$$\sqrt{\frac{I_1}{I_2}} = 2, x+1 = 3x-3$$

$$\frac{I_1}{I_2} = 4, 2x = 4 \Rightarrow x = 2$$

$$I \propto A^2$$

$$A \propto \sqrt{I}$$

$$\frac{A_1}{A_2} = \frac{2}{1}$$

7. If maximum current increases only when intensity increases stopping potential depends on frequency of liquid.
8. λ_x Ray depends on target material do not depend on v_{acc} .

9. Fringe width $\beta = \frac{D\lambda}{d}$

For YDSE -1; $\beta_1 = \frac{D\lambda}{d} = \beta$

For YDSE -2; $\beta_2 = \frac{D\lambda}{2d} = \frac{\beta}{2}$

For YDSE -3; $\beta_3 = \frac{D\lambda}{3d} = \frac{\beta}{3}$

For YDSE -4; $\beta_4 = \frac{D\lambda}{4d} = \frac{\beta}{4}$

10. **For experiment -1**, path difference $\Delta x_1 = \frac{d_1}{D_1} x$

$$\Delta x_1 = \frac{d}{D} \frac{D\lambda}{d} = \lambda$$

$$\therefore \text{Phase difference } \phi_1 = \frac{2\pi}{\lambda} \cdot \lambda = 2\pi$$

$$I_1 = I_0 \cos^2 \frac{\phi_1}{2} = I_0$$

For experiment -2, path difference $\Delta x_2 = \frac{d_2}{D_2} x$

$$\Delta x_2 = \frac{2d}{6D} \frac{D\lambda}{d} = \frac{\lambda}{3}$$

$$\therefore \text{Phase difference } \phi_2 = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{3} = \frac{2\pi}{3}$$

$$I_2 = I_0 \cos^2 \frac{\phi_2}{2} = \frac{I_0}{4}$$

For experiment -3, path difference $\Delta x_3 = \frac{d_3}{D_3} x$

$$\Delta x_3 = \frac{3d}{12D} \frac{D\lambda}{d} = \frac{\lambda}{4}$$

$$\therefore \text{Phase difference } \phi_3 = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$$

$$I_3 = I_0 \cos^2 \frac{\phi_3}{2} = \frac{I_0}{2}$$

For experiment -4, path difference $\Delta x_4 = \frac{d_4}{D_4} x$

$$\Delta x_4 = \frac{4d}{24D} \frac{D\lambda}{d} = \frac{\lambda}{6}$$

$$\therefore \text{Phase difference } \phi_4 = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{6} = \frac{\pi}{3}$$

$$I_4 = I_0 \cos^2 \frac{\phi_4}{2} = \frac{3I_0}{4}$$

11. Find time constant by shorting the battery finding R_{eff} across C_0 . Then $\tau = R_{\text{eff}} \times C_0$.

12. Find time constant by shorting the battery finding C_{eff} across R_0 . Then $\tau = R_0 \times C_{\text{eff}}$.

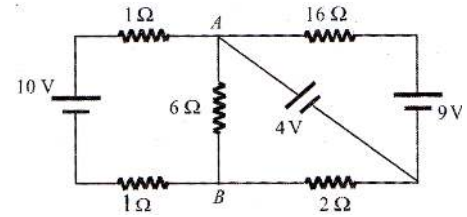
PART - B

1. Potential between A and C is fixed

$$E_{\text{eq}} = \frac{\frac{4}{2} + \frac{10}{2}}{\frac{1}{2} + \frac{1}{2}} = \frac{2+5}{1}$$

$$E_{\text{eq}} = 7$$

$$V_{AB} = 6 \times \left(\frac{7}{7} \right) = 6 \text{ volt}$$



2. $\phi = \frac{q}{6\epsilon_0}$, assume a cube around charge q where one face will be long surface.

3. $\vec{B} = (x\hat{i} + 4\hat{j}) \times 10^{-2}$

$$\vec{a} = \frac{8}{3}\hat{i} - 2\hat{j}$$

$$\vec{F} = q(\vec{V} \times \vec{B})$$

$$\vec{F} \perp \vec{B}$$

$$\vec{a} \cdot \vec{B} = 0$$

$$\frac{8}{3}x - 4 \times 2 = 0$$

$$x = 3$$

4. $F = iLB$
 $F = 48 \times (0.5) (0.25)$
 $F = 6\text{N}$



$$F - f = ma \quad \dots(i)$$

$$f \times R = \frac{mR^2}{2} \alpha \quad \dots(ii)$$

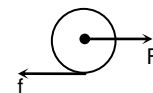
$$f = \frac{ma}{2} \quad \dots(iii)$$

From (i) and (iii)

$$F - \frac{ma}{2} = ma$$

$$F = \frac{3ma}{2}$$

$$f = \frac{ma}{2} = \frac{F}{3}$$



5. $I = \frac{P}{A} = 5$, $F = 10^{-7} \text{ N}$

$$P = 5\text{A}$$

$$F \times V = 3\text{A},$$

$$5\text{A} = 10^{-2} \times 3 \times 10^8$$

$$5\text{A} = 30$$

$$V = C = 3 \times 10^8 \text{ m/s}$$

$$A = \frac{30}{5} = 6 \text{ m}^2$$

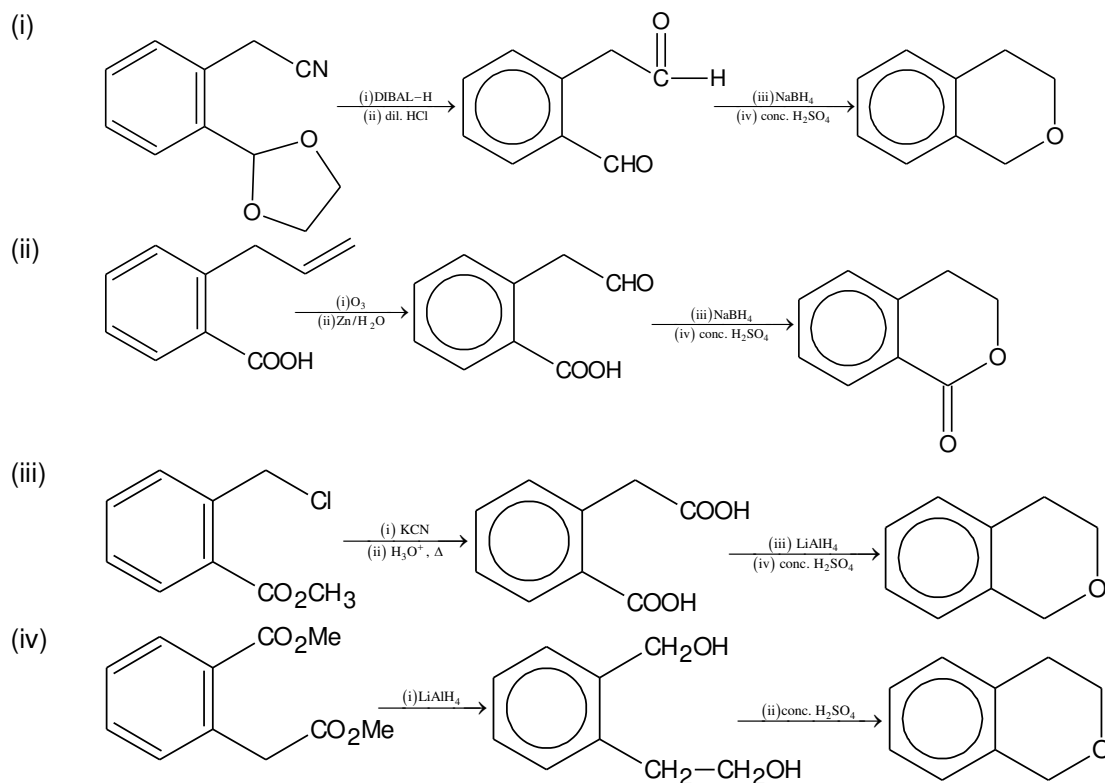
$$6. \quad I = \sqrt{(3-2)^2 + (5)^2} = \sqrt{26} \text{ mA}$$

CHEMISTRY

PART-A

1. $\text{K}_3 \left[\overset{+1}{\text{Cu}}(\text{CN})_4 \right], \overset{+1}{\text{Cu}}\text{SCN}, \overset{+1}{\text{Cu}}_2\text{O}, \left[\overset{+1}{\text{Fe}}(\text{H}_2\text{O})_5\text{NO} \right] \text{SO}_4^{2-}$
2. All carbonyl compounds containing α - H or α - D undergo aldol condensation. (except benzaldehyde)
3. The species having less reduction potential with respect to NO_3^- ($E^0 = +0.96 \text{ V}$) will be oxidised by NO_3^- . These species are V, Fe and Hg.
4. Frenkel defects is also known as dislocation defects.
5. Conceptual
6. (a) CCl_4 and CH_3OH , the extent of H – bonding is decreased. The mixture is more vaporised thus, positive deviation from Raoult's law.
(b) Acetone + CS_2 shows positive deviations.
7. Conceptual
8. Naproxen is a non – steroidal anti inflammatory drug. Tetracycline is used to treat infection. Chloropheniramine is an antihistamine, while equanil is a tranquilizer.

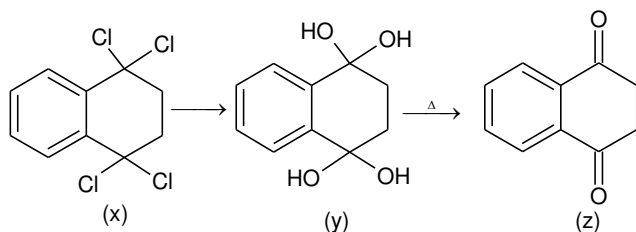
9-10.



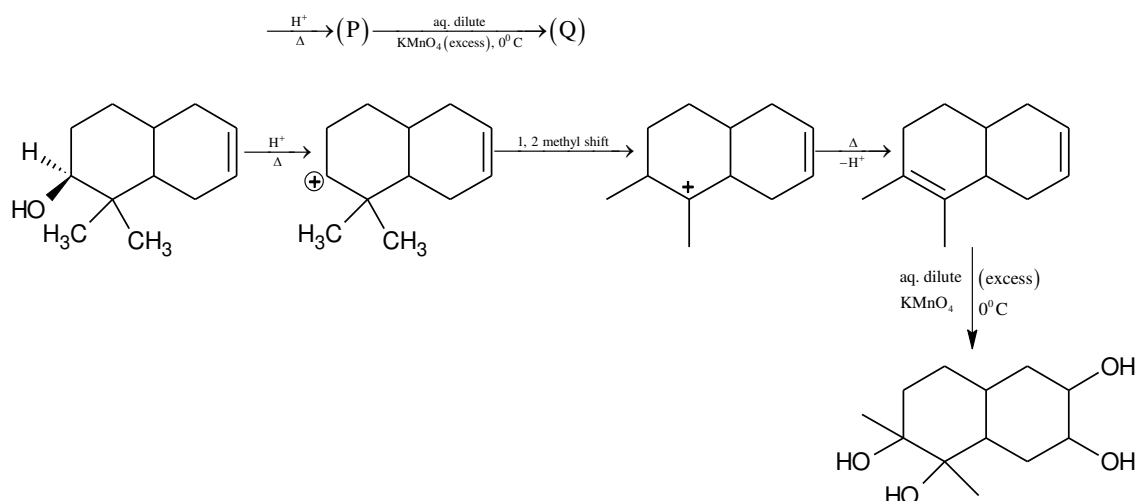
PART-B

1. Benzene is a neutral ligand.

2.

3. It has only one CH_2OH group.

4.



5.

$$A = 2y^{1/3} \text{ nm.} \quad M = 6.023 \times y \times 4 \text{ amu}$$

$$\text{Density} = m/a^3 = \frac{6.023 \times 4 \times 1.66 \times 10^{-27}}{8y \times 10^{-27}} \text{ kg/m}^3$$

$$= \frac{40}{8} = 5 \text{ kg/m}^3$$

6.

$$\text{Mass of benzene} = 33 \times 0.877 \text{ g}$$

$$\text{Moles of benzene} = \frac{28.941}{78} = 0.371$$

$$\text{Mass of hexane} = 187 \times 0.66$$

$$= 123.42 \text{ g}$$

$$m = \frac{0.371 \times 1000}{123.42} = 3.006$$

MATHEMATICS

Part – A

1.

A, B

$$\text{We have } A = AB = A(BA) = (AB)A = AA = A^2$$

$$\text{and } B = BA = B(AB) = (BA)B = BB = B^2$$

2.

A, D

$$R_n = \begin{vmatrix} 3 & 1 & 0 & 0 & \dots \\ 1 & 3 & 1 & 0 & 0 & \dots \\ 0 & 1 & & & \\ & 0 & & & \\ & & & & \end{vmatrix}_{n \times n} \Rightarrow R_n = 3 \begin{vmatrix} 3 & 1 & 0 & \dots \\ 1 & & & \\ 0 & & & \\ & & & \end{vmatrix}_{(n-1) \times (n-1)} + \begin{vmatrix} 1 & 1 & 0 & 0 & \dots \\ 0 & 3 & & & \\ 0 & 1 & & & \\ & 0 & & & \\ & & & & \end{vmatrix}_{(n-1) \times (n-1)}$$

$$\Rightarrow R_n = 3R_{n-1} - R_{n-2}$$

$$\Rightarrow R_n + R_{n-2} = 3R_{n-1}$$

3.

A, B, C

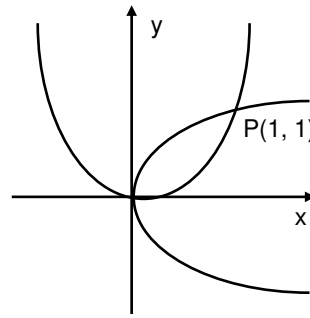
Point of intersection of two curves $y = x^2$ and $x = y^2$ is $(1, 1)$.

Now area bounded by curves is given by

$$A = \int_0^1 (\sqrt{x} - x^2) dx = \left(\frac{2}{3} x^{3/2} - \frac{x^3}{3} \right)_0^1$$

$$\therefore A = \frac{1}{3} \text{ sq. units.}$$

Also, option (A) represents the same region.



4. **B, C**

$$\int (2y + \cos y) dy = \int 6x^2 dx \Rightarrow y^2 + \sin y = 2x^3 + c$$

$$\therefore y(1) = \pi \Rightarrow c = \pi^2 - 2$$

$$\therefore \text{solution is } y^2 + \sin y = 2x^3 + \pi^2 - 2.$$

5. **A, C**

$$(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = \frac{\vec{b} + \vec{c}}{2}$$

$$\vec{b} \left(\vec{a} \cdot \vec{c} - \frac{1}{2} \right) - \vec{c} \left(\vec{a} \cdot \vec{b} + \frac{1}{2} \right) = 0$$

$$\Rightarrow \vec{a} \cdot \vec{c} = \frac{1}{2} \text{ and } \vec{a} \cdot \vec{b} = -\frac{1}{2}$$

$$\Rightarrow \theta_1 \text{ is } \frac{\pi}{3} \text{ and } \theta_2 = 120^\circ$$

6. **A, C**

Since point of intersection of the given lines is $(0, 0, 0)$. It must lie on the angle bisector so,

$$\frac{0-2}{-1} = \frac{0+2}{1} = \frac{0+k^2-1}{4} \Rightarrow k = \pm 3.$$

7. **A, D**

$$P\left(\frac{X}{Y}\right) = \frac{P(X \cap Y)}{P(Y)} = \frac{1}{2}$$

$$P\left(\frac{Y}{X}\right) = \frac{P(X \cap Y)}{P(X)} = \frac{2}{5}$$

$$\Rightarrow P(X \cap Y) = \frac{2}{15}$$

$$P(Y) = \frac{4}{15} \text{ also } P(X \cup Y) = \frac{7}{15}$$

$$P\left(\frac{X'}{Y}\right) = 1 - P\left(\frac{X}{Y}\right) = 1 - \frac{1}{2} = \frac{1}{2}$$

8. **B, D**

$$\sum_{k=1}^{98} \int_k^{k+1} \frac{1}{x+1} dx < \sum_{k=1}^{98} \int_k^{k+1} \frac{k+1}{x(x+1)} dx < \sum_{k=1}^{98} \int_k^{k+1} \frac{dx}{x}$$

$$\sum_{k=1}^{98} (\ln(k+2) - \ln(k+1)) < I < \sum_{k=1}^{98} (\ln(k+1) - \ln k)$$

$$\ln 50 < I < \ln 99$$

$$\Rightarrow \frac{49}{50} < I < \ln 99.$$

9. **A**

10. **B**

9-10. Plane through line $y = ax, z = c$ is $(y - ax) + \lambda_1(z - c) = 0$

$$\text{Similarly, } (y + ax) + \lambda_2(z + c) = 0$$

$$\Rightarrow (y + c) + \lambda_1(y - c) = 0 \text{ and } (y - c) + \lambda_2(y + c) = 0$$

$$\Rightarrow -\lambda_1 = \frac{y+c}{y-c}, \quad \frac{-1}{\lambda_2} = \frac{y+c}{y-c}; \lambda_1 \lambda_2 = 1$$

$$\therefore \text{Locus is } \left(\frac{y-ax}{z-c} \right) \left(\frac{y+ax}{z+c} \right) = 1$$

$$a^2x^2 - y^2 + z^2 - c^2 = 0$$

11. **C**12. **A**

$$11-12. f(x) = 0 \Rightarrow \sin(\pi \cos x) = 0$$

$$\Rightarrow \pi \cos x = n_1 \pi, n_1 \in \mathbb{I}$$

$$\Rightarrow \cos x = -1, \cos x = 0, \cos x = 1$$

$$\Rightarrow x = \frac{n_2 \pi}{2}, n_2 \in \mathbb{I}$$

$$\Rightarrow X = \left\{ \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \dots \right\}$$

$$f'(x) = 0 \Rightarrow -\cos(\pi \cos x) \pi \sin x = 0$$

$$\Rightarrow \sin x = 0 \text{ or } \cos(\pi \cos x) = 0$$

$$\Rightarrow x = n_3 \pi \text{ or } \pi \cos x = (2n_4 + 1) \pi/2$$

$$\Rightarrow \cos x = -1/2, 1/2$$

$$\Rightarrow x = n_5 \pi \pm \pi/3$$

$$Y = \left\{ \frac{\pi}{3}, \frac{2\pi}{3}, \frac{3\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{6\pi}{3}, \dots \right\}$$

$$g(x) = 0 \Rightarrow \cos(2\pi \sin x) = 0$$

$$\Rightarrow 2\pi \sin x = (2n_6 + 1) \frac{\pi}{2}$$

$$\Rightarrow \sin x = \frac{2n_6 + 1}{4} = -\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}$$

$$\Rightarrow Z = \left\{ x \mid \sin x = \pm \frac{1}{4}, \pm \frac{3}{4}, x > 0 \right\}$$

$$g'(x) = 0 \Rightarrow -2\pi \cos x \sin(2\pi \sin x) = 0$$

$$\cos x = 0 \text{ or } \sin(2\pi \sin x) = 0$$

$$\Rightarrow x = (2n_7 + 1) \frac{\pi}{2} \text{ or } 2\pi \sin x = n_8 \pi$$

$$\Rightarrow \sin x = -1, -\frac{1}{2}, 0, \frac{1}{2}, 1$$

$$W = \left\{ x \mid \sin x = \pm 1, \pm \frac{1}{2}, 0, x > 0 \right\}$$

$$\Rightarrow \text{I} - (P), (Q)$$

$$\text{II} - (Q), (T)$$

$$\text{III} - (R)$$

$$\text{IV} - (P), (R), (S)$$

Part – B1. **1.00**

$$\text{Det}(z) = 0 \times 0 - 1 \times 1 = 0 - 1 = -1$$

$$p = \text{Det}(\text{Det}(z)) = (-1)^2 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$\Rightarrow |p| = 1$$

2. **3.00**

Let the circumcentre of the triangle be the origin.

$$\Rightarrow \text{orthocentre is } \vec{a} + \vec{b} + \vec{c} \text{ and the centroid is } \frac{\vec{a} + \vec{b} + \vec{c}}{3} \Rightarrow F = \frac{2}{3}(\vec{a} + \vec{b} + \vec{c})$$

$$\Rightarrow (\vec{a} - \vec{F})^2 + (\vec{b} - \vec{F})^2 + (\vec{c} - \vec{F})^2 = \frac{1}{9} [(\vec{a} - 2(\vec{b} + \vec{c}))^2 + (\vec{b} - 2(\vec{a} + \vec{c}))^2 + (\vec{c} - 2(\vec{a} + \vec{b}))^2]$$

$$= \frac{1}{9}(27) = 3.$$

3. **1.00**

The lines should intersect or be parallel

Let $(2t + 1, -t, 2t)$ be any point on the first linePutting it in both the planes we get $\lambda = \frac{-31}{7}$.There is no λ for which both the lines are parallel.4. **0.40**

Initially 4 Red balls and 6-Black balls

$$\text{Required probability} = \frac{4}{10} \times \frac{6}{12} + \frac{6}{10} \times \frac{4}{12} = \frac{2}{5}$$

5. **0.20**Given set of numbers is $\{1, 2, 3, 4, 5, 6, 7, 8\}$

We have to select three numbers

Let us assume that

event $A \rightarrow$ in selected three numbers minimum is 3event $B \rightarrow$ in selected three numbers maximum is 6We have to find $P(A/B)$

$$\text{Now } P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{n(B)}$$

 $A \cap B$ = in selected three numbers, minimum is 3 and maximum is 6 $\therefore$ The remaining number is either 4 or 5 $\therefore n(A \cap B) = 2$ $n(B)$ = selecting two numbers from 1, 2, 3, 4, 5 (as maximum of numbers is 6)
 $= {}^5C_2$

$$\therefore P(A/B) = \frac{n(A \cap B)}{n(B)} = \frac{2}{{}^5C_2} = \frac{1}{5}.$$

6. **2.00**

$$g(x) = \int_x^{\pi/2} [f'(t) \operatorname{cosec} t - \cot(t) \operatorname{cosec}(t) f(t)] dt$$

$$\Rightarrow g(x) = f(t) \operatorname{cosec}(t) \Big|_x^{\pi/2} = 3 - \frac{f(x)}{\sin x}$$

$$\Rightarrow \lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} \left(3 - \frac{f(x)}{\sin x} \right)$$

$$= 3 - f'(0) = 2$$